MAT21-4

Discrete Stochastic Models & Concentration

Conditional expectation, random walks, Markov chains, concentration inequalities

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Structured stochastic reasoning and quantitative uncertainty bounds on finite and countable spaces.

Albert School — 22 hours

Contents

1 Conditional expectation on finite partitions

We work on a finite probability space (Ω, P) and let $X : \Omega \to \mathbb{R}$ be a random variable, where $\mathbb{R}$ is written $\mathbb{R}$. A **partition** of Ω is a family $\mathcal{P} = \{B_1, ..., B_k\}$ of disjoint events with $\bigcup_i B_i = \Omega$ and $P(B_i) > 0$. A partition encodes the information "which block B_i occurred".

Definition 1 (Conditional expectation given a block) For an event B with $P(B) > 0$,

$$E[X \mid B] = \sum_x x\, P(X = x \mid B) = \frac{1}{P(B)} \sum_{\omega \in B} X(\omega)\, P(\omega).$$

Definition 2 (Conditional expectation given a partition) The conditional expectation of X given $\mathcal{P}$ is the random variable $E[X \mid \mathcal{P}]$ that takes the constant value $E[X \mid B_i]$ on every outcome of B_i:

$$E[X \mid \mathcal{P}](\omega) = E[X \mid B_i] \quad \text{if } \omega \in B_i.$$

Computation proceeds by **partition refinement**: to compute $E[X \mid \mathcal{P}]$ one restricts to each block B_i and averages X over that block.

1.1 The tower property

A partition $\mathcal{Q}$ **refines** $\mathcal{P}$ if every block of $\mathcal{Q}$ is contained in a block of $\mathcal{P}$; then $\mathcal{P}$ is **coarser** and carries less information.

Theorem 3 (Tower property) If $\mathcal{Q}$ refines $\mathcal{P}$, then

$$E[E[X \mid \mathcal{Q}] \mid \mathcal{P}] = E[X \mid \mathcal{P}].$$

In particular, taking $\mathcal{P}$ to be the trivial partition $\{\Omega\}$,

$$E[E[X \mid \mathcal{Q}]] = E[X].$$

Conditioning on a finer partition is "information gain": each refinement step can only sharpen the conditional expectation, and re-coarsening recovers the previous value.

> **Worked example — Verifying the tower property.** Roll a fair die, $\Omega = \{1, ..., 6\}$, $X(\omega) = \omega$. Let $\mathcal{Q}$ be the finest partition into singletons and $\mathcal{P} = \{\{1, 2, 3\}, \{4, 5, 6\}\}$ (low/high). Then $E[X \mid \mathcal{P}]$ equals 2 on $\{1, 2, 3\}$ and 5 on $\{4, 5, 6\}$, and $E[E[X \mid \mathcal{Q}] \mid \mathcal{P}] = E[X \mid \mathcal{P}]$ since $E[X \mid \mathcal{Q}] = X$. Averaging gives $E[X \mid \mathcal{P}]$ with mean $\frac{1}{2}(2) + \frac{1}{2}(5) = 3.5 = E[X]$.

2 Variance decomposition

Theorem 4 (Law of total expectation) For a partition $\mathcal{P} = \{B_1, ..., B_k\}$,

$$E[X] = \sum_{i=1}^{k} E[X \mid B_i]\, P(B_i) = E[E[X \mid \mathcal{P}]].$$

Theorem 5 (Law of total variance)

$$\mathrm{Var}(X) = \underbrace{E[\mathrm{Var}(X \mid \mathcal{P})]}_{\text{residual (unexplained)}} + \underbrace{\mathrm{Var}(E[X \mid \mathcal{P}])}_{\text{explained by } \mathcal{P}}.$$

Here $\mathrm{Var}(X \mid \mathcal{P})$ is the random variable equal to the within-block variance $\mathrm{Var}(X \mid B_i)$ on B_i. The decomposition splits total variability into the part **explained** by knowing which block occurred and the **residual** variability remaining inside the blocks.

> **Worked example — Decomposing variance by conditioning.** Pick a coin uniformly from two coins, one with heads-probability 0.2 and one with 0.8, then flip it once; let $X \in \{0, 1\}$ be the result and $\mathcal{P}$ the partition by which coin was chosen. Each block has within-block variance $0.2 \cdot 0.8 = 0.16$, so the residual term is 0.16. The block means are 0.2 and 0.8 with overall mean 0.5, so the explained term is $\frac{1}{2}(0.2 - 0.5)^2 + \frac{1}{2}(0.8 - 0.5)^2 = 0.09$. Total $\mathrm{Var}(X) = 0.25$.

3 Martingale intuition

We use martingales as the formalisation of a **fair game**, without any filtration apparatus.

Definition 6 (Martingale (fair game)) A sequence $X_0, X_1, X_2, ...$ is a martingale if for every n,

$$E[X_{n+1} \mid X_0, ..., X_n] = X_n.$$

The conditional expected next value equals the current value: on average the game neither gains nor loses.

Canonical examples:

- **Symmetric random walk:** $S_n = \sum_{i=1}^{n} \xi_i$ with $\xi_i = \pm 1$ equally likely. Since $E[\xi_{n+1}] = 0$, $E[S_{n+1} \mid S_0, ..., S_n] = S_n$.
- **Doob martingale:** fix a random variable Y and a refining sequence of partitions; $X_n = E[Y \mid \mathcal{P}_n]$ is a martingale by the tower property.

3.1 Optional stopping

A **stopping time** T is a random time whose occurrence "by time n" depends only on $X_0, ..., X_n$. The fair-game intuition suggests $E[X_T] = E[X_0]$, i.e. no stopping rule turns a fair game into a favourable one.

Theorem 7 (Optional stopping (conditions)) If (X_n) is a martingale and T a stopping time, then $E[X_T] = E[X_0]$ provided one of the following holds:

1. T is bounded ($T \le N$ for a constant N); or
2. the increments are bounded and $E[T] < \infty$; or
3. $X_{n \wedge T}$ is dominated by an integrable random variable.

Common pitfall Without such a condition the conclusion can fail.

Worked example — A counterexample to optional stopping. Take the symmetric random walk S_n (a martingale, $S_0 = 0$) and the doubling ("martingale") betting strategy: keep playing until the first win, $T = \inf\{n : S_n = 1\}$. The walk is recurrent, so $T < \infty$ almost surely and $S_T = 1$, giving $E[S_T] = 1 \ne 0 = E[S_0]$. Optional stopping fails because $E[T] = \infty$: none of the conditions above hold.

4 Random walks

Let $S_0 = k$ and $S_n = k + \sum_{i=1}^{n} \xi_i$ with independent steps $\xi_i = +1$ with probability p and $\xi_i = -1$ with probability $q = 1 - p$. The walk is **symmetric** if $p = \frac{1}{2}$ and **biased** otherwise.

4.1 Return probabilities

The symmetric walk on $\mathbb{Z}$ is **recurrent**: starting from 0 it returns to 0 with probability 1. The biased walk ($p \ne \frac{1}{2}$) is **transient**: it returns to its start with probability < 1 and drifts to $\pm\infty$.

4.2 Gambler's ruin

A gambler starts with capital k, plays unit bets, and stops on reaching either 0 (ruin) or N (target); the states 0 and N are **absorbing**. Let h_k be the probability of reaching N before 0.

Proposition 8 (Ruin probabilities) With $r = \frac{q}{p}$,

$$h_k = \begin{cases} \frac{k}{N} & \text{if } p = \frac{1}{2} \\ \frac{r^k - 1}{r^N - 1} & \text{if } p \ne \frac{1}{2}. \end{cases}$$

4.3 Hitting times

Let τ_k be the expected number of steps until absorption, starting from k.

Proposition 9 (Expected duration (symmetric case)) For $p = \frac{1}{2}$,

$$E[\tau_k] = k(N - k).$$

These quantities are obtained by **first-step analysis**: conditioning on the first step gives a recurrence $h_k = p\, h_{k+1} + q\, h_{k-1}$ (with $h_0 = 0$, $h_N = 1$), and similarly for the expected duration.

5 Markov chains

Definition 10 (Markov chain and transition matrix) A sequence (X_n) on a finite or countable state space S is a Markov chain if

$$P(X_{n+1} = j \mid X_n = i, ..., X_0) = P(X_{n+1} = j \mid X_n = i) =: P_{ij}.$$

The **transition matrix** $P = \left(P_{ij}\right)$ has nonnegative entries and rows summing to 1. The n-step transition probabilities are the entries of P^n.

5.1 Classification of states

- A state i is **absorbing** if $P_{ii} = 1$.
- i is **recurrent** if, started from i, the chain returns to i with probability 1; otherwise it is **transient**.
- The chain is **irreducible** if every state can be reached from every other state.
- The **period** of i is $\gcd\left\{n \geq 1 : (P^n)_{ii} > 0\right\}$; the chain is **aperiodic** if every state has period 1.

5.2 Stationary distributions

Definition 11 (Stationary distribution) A row vector $\pi = (\pi_i)$ with $\pi_i \geq 0$ and $\sum_i \pi_i = 1$ is a stationary distribution if

$$\pi P = \pi.$$

For an irreducible aperiodic chain on a finite state space the stationary distribution exists, is unique, and is the limit of the row distributions: $P(X_n = j) \to \pi_j$ as $n \to \infty$.

> **Worked example — A two-state chain.** With $P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$, solving $\pi P = \pi$ with $\pi_1 + \pi_2 = 1$ gives $\pi = \left(\frac{4}{7}, \frac{3}{7}\right)$. The chain is irreducible and aperiodic, so the distribution of X_n converges to π.

6 Concentration inequalities

Concentration inequalities bound the probability that a random variable deviates from its mean, turning variability into **quantitative** tail bounds.

Theorem 12 (Markov's inequality) If $X \geq 0$ and $a > 0$, then

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

Proof. Since $X \geq 0$, $a\,\mathbf{1}_{\{X \geq a\}} \leq X$ pointwise. Taking expectations, $a\,P(X \geq a) \leq E[X]$, and dividing by a gives the claim. □

Theorem 13 (Chebyshev's inequality) If X has mean μ and finite variance σ^2, then for $k > 0$,

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

Proof. Apply Markov's inequality to the nonnegative variable $(X-\mu)^2$ with threshold k^2: $P(|X-\mu| \geq k) = P\big((X-\mu)^2 \geq k^2\big) \leq \frac{E[(X-\mu)^2]}{k^2} = \frac{\sigma^2}{k^2}$. $\square$

Theorem 14 (Hoeffding's bound) Let $X_1, ..., X_n$ be independent with $a_i \leq X_i \leq b_i$, and $S = \sum_{i=1}^{n} X_i$. Then for $t > 0$,

$$P(S - E[S] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^{n} (b_i - a_i)^2}\right),$$

and the same bound holds for $P(S - E[S] \leq -t)$.

The Hoeffding and Chernoff bounds give **exponentially** decaying tail estimates, far sharper than Chebyshev for sums of bounded independent variables. The Chernoff method bounds $P(S \geq a) \leq e^{-\lambda a} E\left[e^{\lambda S}\right]$ and optimises over $\lambda > 0$; Hoeffding's bound is obtained this way. These are **verified empirically in Python** (see the mini-labs) by comparing the theoretical bound to observed tail frequencies.

7 Quantitative uncertainty control

Concentration bounds translate directly into **sample-size requirements**. Let $X_1, ..., X_n$ be independent with values in $[0, 1]$, mean μ, and sample mean $\bar{X} = \frac{1}{n} \sum_i X_i$. Hoeffding's bound gives

$$P\big(|\bar{X} - \mu| \geq \varepsilon\big) \leq 2\exp(-2n\varepsilon^2).$$

Proposition 15 (Minimum sample size) To guarantee $P\big(|\bar{X} - \mu| \geq \varepsilon\big) \leq \delta$ it suffices to take

$$n \geq \frac{1}{2\varepsilon^2} \ln\left(\frac{2}{\delta}\right).$$

This is the standard "confidence level $1 - \delta$, error tolerance ε" calculation: it returns the minimum number of observations needed to make a reliability statement of the desired strength.

Worked example — A sample-size computation. For error tolerance $\varepsilon = 0.05$ and confidence $1 - \delta = 0.95$ ($\delta = 0.05$),

$$n \geq \frac{1}{2 \cdot 0.05^2} \ln\left(\frac{2}{0.05}\right) = 200 \ln(40) \approx 737.$$

8 Python mini-labs

The labs implement the models above and compare theory to simulation.

Exercises

1. Simulate a random walk. Generate ± 1 steps and accumulate S_n; estimate the gambler's-ruin probability h_k from 0 to N by Monte Carlo and compare to the closed form k/N (symmetric) or $\big(r^k - 1\big)/\big(r^N - 1\big)$ (biased).

2. Empirically verify concentration. For a sum S of n independent bounded variables, estimate $P(S - E[S] \geq t)$ by repeated simulation and overlay the observed tail frequency on Hoeffding's bound $\exp\left(-2t^2 / \sum (b_i - a_i)^2\right)$.
3. Compare bounds to observed tails. For a fixed deviation, tabulate the Chebyshev bound, the Hoeffding bound, and the simulated tail probability, and confirm that Hoeffding is exponentially tighter while both remain valid upper bounds.
4. Estimate a martingale. Simulate the symmetric random walk as a martingale, estimate $E[S_T]$ for a bounded stopping time $T \leq N$ and check $E[S_T] \approx S_0$; then take the unbounded doubling-strategy stopping time and observe the failure of optional stopping.